

Ayush Singh

+91 6386846969 | ayush0987singh@gmail.com | [LinkedIn](#) | [Portfolio](#) | [GitHub](#) | [Codeforces](#) | [LeetCode](#)

EDUCATION

Vellore Institute of Technology, Vellore

Masters of Computer Applications

Jun. 2025 – Present

CGPA: 9.42

University of Lucknow

Bachelors of Computer Applications

Oct. 2022 – 2025

Percentage: 75.14%

Indian Institute of Technology Madras

BS in Data Science and Programming [Distant Mode]

Sept. 2021 – 2025

CGPA: 7.42

Army Public School SP Marg Lucknow

Senior Secondary Education

Jun. 2019 – April 2021

Percentage: 95.20%

- Awarded Academic Excellence; served as House Captain

PROJECTS

AI-Powered Duplicate File Detection System | *Python, OpenAI CLIP, SBERT, FAISS, OpenCV, librosa*

- * Collaborated in a team of 3 to build a multi-modal deduplication engine for images, documents, audio, and video using CLIP embeddings, SBERT semantic search, and Chromaprint fingerprinting.
- * Designed a layered pipeline (SHA-256 → perceptual hashing → neural embeddings) reducing neural inference calls by ~70% via fast pre-filtering.
- * Integrated FAISS ANN search enabling real-time similarity queries across 100k+ files in sub-linear time; packaged as a deployable AI desktop application with non-destructive OS trash integration.

Payment Processing Microservice | *Java, Spring Boot, PostgreSQL, Redis, RabbitMQ, REST APIs*

- * Built an Idempotency Engine using Redis ensuring every payment request is processed exactly once; implemented Saga Pattern across Order, Payment, and Inventory services.
- * Integrated Circuit Breaker (Resilience4j) for gateway downtime handling and Webhook Handler with retry logic for async Stripe/PayPal notifications.
- * Maintained immutable Audit Logs per transaction state change with PCI-DSS-aware tokenization and input validation.

EV Demand Forecasting | *Python, TensorFlow, Scikit-Learn, Pandas*

- * Engineered a time-series data pipeline with data preprocessing and feature engineering on macroeconomic and EV pricing datasets; designed a hybrid CNN-LSTM-Attention model with a custom Attention layer from tensor operations.
- * Validated robustness via 10-fold cross-validation and hyperparameter optimization, improving prediction accuracy over baseline LSTM.

Local RAG QA System with Multi-Query Retrieval | *Python, LangChain, ChromaDB, Ollama, LLaMA 3.2*

- * Built an offline RAG pipeline using ChromaDB for low-latency semantic retrieval of domain documents.
- * Implemented LangChain Multi-Query Retriever to improve context recall over standard similarity search.

SKILLS

Languages : Java, Python, SQL, C++

Backend & APIs : Spring Boot, FastAPI, RESTful APIs, Microservices Architecture

ML & AI : PyTorch, TensorFlow, Scikit-Learn libraries, Data Preprocessing, XGBoost, LangChain, ChromaDB, SBERT

Databases : PostgreSQL, MySQL, SQLite3, MongoDB, Redis; ACID Transactions, Indexing

Cloud & DevOps : Azure, GCP, Docker, Kubernetes, Git, Postman

Core Concepts : OOP, Multithreading, Distributed Systems, Linux/Unix, Computer Networks (TCP/IP, HTTP, DNS)

Frontend (Basic) : React.js, HTML, CSS

ACHIEVEMENTS & PROFILES

- Solved **382+ LeetCode problems** across arrays, trees, graphs, dynamic programming, and system design.
- Earned **GCP Arcade Legend** on Google Cloud Skills Boost through hands-on labs in cloud infra and data services.
- Received **Academic Excellence** at Army Public School; served as **House Captain**, leading teams and events.

RELEVANT COURSEWORK

Core CS: Data Structures & Algorithms, Operating Systems, DBMS, Computer Networks, OOP, System Design

Advanced & Systems: Distributed Systems, Cloud Computing, Machine Learning, Deep Learning, Software Engineering, Information Security